

Proposal: Digitizing the PDM Final Project Lifecycle

1. Overview

Item	Details
Working Project Title	Digitization of the PDM Final Project Lifecycle
Project Type	Product Discovery, Definition, Design and Validation
Team Size	2 Students
Start Date	14 August 2026
Planned Duration	20 Weeks
Weekly Commitment	4 Hours per Student
Individual Contribution	80 Hours per Student
Total Team Effort	320 Person-Hours
Primary Context	PDM Final Projects
Primary Stakeholders	Students / Project Teams, Faculty Mentors
Programme Stakeholder	PDM Programme Coordinator / Relevant Programme Stakeholder
Committed Final Outcome	Research-backed and validated interactive prototype of the proposed MVP, supported by product requirements, research findings and development handoff
Production Software Development	Not committed within the initial scope
Potential Continuation	Future student team, development team or faculty-led implementation

2. Context

PDM final projects involve students working individually or in teams under the guidance of faculty mentors over an extended project lifecycle.

Based on an initial discussion with faculty, several activities related to final projects currently appear to be handled manually or through different tools and communication channels.

The initial discussion also raised the possibility that mentors may have strong visibility into the projects they directly supervise but limited visibility into projects being handled under other mentors or into the overall progress of projects across a batch.

However, these observations are currently only **initial inputs and assumptions**.

No structured primary research has yet been conducted with students, mentors or programme stakeholders.

Therefore, this proposal does not assume that a particular software solution or predefined feature set is already required.

The project will first study the existing final-project lifecycle, validate the actual problems and then determine where digital intervention can create meaningful value.

3. Initial Context to Be Investigated

Area	Initial Observation / Question
Project Initiation	How are projects proposed, approved, assigned and started?
Mentor Allocation	How are mentors selected or assigned to students or teams?
Project Progress	How is progress currently tracked and communicated?
Meetings	How are mentor-student discussions currently managed?
Minutes of Meeting	Are MOMs created, stored and referred to later?
Action Items	How are actions assigned, tracked and carried forward?
Deliverables	How are milestones, submissions and project outputs managed?
Documentation	Where is project documentation maintained throughout the lifecycle?

Mentor Visibility	What information does a mentor need about their own projects and, if useful, other projects?
Programme Visibility	Does anyone currently have a complete view of project progress across a batch?
Reviews	How are reviews conducted and how are decisions recorded?
Completion	What formally defines project completion?
Archival	What information is preserved once a project is completed?
Continuity	Can future students discover, learn from or continue previous projects?

These areas are starting points for research and should not be interpreted as validated problems.

4. Proposed Project Direction

We propose to undertake a structured product discovery, definition, design and validation project focused on the **PDM Final Project Lifecycle**.

The team will study how final projects currently operate from initiation to completion before determining what should be digitized.

The proposed progression is:

Stage	Core Question
Understand	How does the current final-project system work?
Research	What problems are actually experienced by students, mentors and programme stakeholders?
Define	Which problems are important enough to solve?
Prioritize	Which opportunity provides the highest value within the available scope?
Design	What should the proposed product experience look like?
Prototype	Can the proposed workflow be represented and experienced interactively?
Validate	Does the concept work for representative users?
Iterate	What needs to change based on evidence?
Handoff	What should be built next if the product is taken forward?

5. Project Goal

The broad working question for the project is:

How might the PDM final-project lifecycle be improved through appropriate digital interventions so that students can manage and document their work effectively, mentors can guide and monitor projects with better context, and the programme can preserve useful project knowledge across batches?

This is currently a **working problem space**, not the final problem statement.

The final problem statement will emerge after primary research and synthesis.

6. Project Objectives

No.	Objective
1	Understand and document the existing PDM final-project lifecycle.
2	Study the experiences and needs of students, project teams, faculty mentors and relevant programme stakeholders.
3	Identify problems, inefficiencies and opportunities within the current process.
4	Prioritize problems that may benefit from digital intervention.
5	Define the requirements and scope of a potential digital product.
6	Design an interactive prototype representing the proposed MVP.
7	Test the prototype with representative users.
8	Iterate the proposed product based on validation findings.
9	Produce a structured product and development handoff that can be taken forward for future implementation.

7. Why Full Development Is Not Being Committed Upfront

The actual user problems, workflows, product requirements and system boundaries have not yet been validated.

Beginning production software development before answering these questions would require the team to make assumptions regarding:

Unknown	Why It Matters
Actual user problems	The team may solve a problem that users do not consider important.
Stakeholder roles	The system may provide incorrect or unnecessary access.
Project lifecycle	Different project types may follow different processes.
Information requirements	The team may track information nobody needs.
Existing tools	A new product may unnecessarily duplicate Moodle, Drive or other systems.
Visibility requirements	Cross-project visibility may be useful, unnecessary or sensitive.
Archival needs	It is not yet known what information should survive after project completion.
Feature priority	Development effort may go into low-value functionality.

Therefore, the project will follow:

Understand → Research → Define → Design → Prototype → Validate → Iterate → Handoff

A full production implementation is currently outside the committed scope.

If research identifies a sufficiently small opportunity that can realistically be implemented within the available capacity, a lightweight proof of concept or low-code implementation may be explored.

However, production development will not be required for successful completion of the proposed fellowship work.

8. Stakeholders

Stakeholder	Role in Final Projects	What the Research Needs to Understand
Students / Project Teams	Carry out projects, work with mentors, manage project work and produce deliverables	How they organize work, document progress, manage meetings, actions, deadlines and artefacts
Faculty Mentors	Guide projects, review progress, provide feedback and support decisions	How they monitor projects, conduct reviews and what information they need
PDM Programme Coordinator / Programme Stakeholder	May manage batches, projects, mentors, reviews, completion and archival	What programme-level visibility, administration and reporting are required
Previous Project Students	Have completed the entire project lifecycle	What worked, what was lost and what they wish had been preserved
Future Batches	Potential users of archived project knowledge	Whether previous project information can support future work
Other Stakeholders	May include reviewers, external mentors or administrators	To be identified through research

Additional stakeholders will only be included where research establishes their relevance.

9. Initial Research Areas

9.1 Project Initiation

Research Question
How does a final project formally begin?
How are individual students or project groups formed?
How are project topics selected?
How are mentors selected or assigned?
What information needs to be submitted at the beginning?

Who approves a project?
Is there a formal project proposal or initiation process?

9.2 Project Lifecycle

Research Question
What stages does a final project currently go through?
Are stages formally defined?
Do different types of projects follow different processes?
What milestones exist?
What reviews are compulsory?
What determines whether a project can move forward?

9.3 Student Project Management

Research Question
How do students currently manage their project?
Which tools are currently being used?
How are tasks and action items managed?
How are deadlines tracked?
How are project decisions documented?
How is team responsibility distributed?
Where does information become fragmented or difficult to retrieve?

9.4 Student–Mentor Interaction

Research Question
How frequently do students meet mentors?
How are meetings scheduled?
Are Minutes of Meeting currently documented?
Who creates them?

Where are they stored?
How are decisions from meetings recorded?
How are action items assigned?
How does a mentor check whether previous action items were completed?
How is feedback communicated?

9.5 Mentor Workflow

Research Question
How many projects does a mentor typically supervise?
How does a mentor currently monitor project progress?
What information does a mentor need before a review?
What information do mentors repeatedly ask students to provide?
How does a mentor identify projects that may be falling behind?
What information should a mentor be able to access about other projects?
Is visibility across projects and mentors actually required?
What information should remain private?

9.6 Programme-Level Visibility

Research Question
Who currently has visibility into all final projects?
How is batch-level progress monitored?
Is there a need to compare projects?
Is there a need to know which stage different projects are in?
How are delayed or inactive projects currently identified?
What programme-level reporting is required?

9.7 Documents and Artefacts

Research Question
What documents are generated throughout a project?
Where are these documents currently stored?
Which artefacts are mandatory?
Which artefacts are useful but optional?
Are project documents distributed across Drive, email, WhatsApp, Moodle, GitHub or other systems?
Who owns the final project files?

9.8 Project Completion

Research Question
What constitutes completion of a final project?
What final submissions are required?
Who approves completion?
What happens to unfinished projects?
What happens when a student graduates?

9.9 Project Archive and Continuity

Research Question
How are completed projects currently archived?
What information is retained?
What information gets lost?
Can students access previous projects?
Can future batches discover earlier work?
Are previous projects ever continued by another student or batch?
Would retaining project-process knowledge provide value in addition to retaining the final project outcome?

10. Existing PDM Final Project Archive

The PDM programme already maintains information about projects completed by previous batches.

Examples currently available include:

Unlabelled.FM (Jan 2022), Fytt.io (Jan 2022), Finsocial (Jan 2022), For I-Dx (Jan 2022), Aurora (Jan 2022), ScientiQuick (Jan 2022), Flowsight (Jan 2022), Kyou2 (July 2022), GreenTrace (July 2022), Finease (July 2022), MediSync (July 2022), VR Resilience Builder (July 2022), APPRAISE (July 2022), ConceptCure (July 2022), JobSims (July 2022), Pocket Pal (July 2022), Distro-Cred (July 2022), Intelli-Dispute (July 2022), SQUAD (July 2022), Englingo (July 2022), Paddy Weeder (July 2022), Safe Travel India (July 2022).

The existing archive contains information such as:

Existing Information
Project title
Product / solution summary
Student / team members
Batch information
Project poster information

The archive demonstrates that final projects differ significantly across:

- domains,
- solution types,
- team sizes,
- levels of technical complexity,
- and intended outcomes.

Therefore, the eventual system should not automatically assume that all PDM final projects follow an identical project-development process.

The archive will be treated as an **input into the research** and may later provide realistic sample historical data for prototype testing.

It will **not** be treated as evidence of a specific user problem.

11. Proposed Outcomes

Outcome Area	Expected Result
Current-State Understanding	Clear view of how final projects currently operate
Research	Evidence from students, previous project students, mentors and programme stakeholders
Problem Definition	Validated and prioritized problem statement
Product Definition	Defined users, roles, requirements, scope and MVP
Experience Design	Information architecture, journeys, workflows and wireframes
Prototype	High-fidelity interactive representation of the proposed MVP
Validation	Usability-testing findings and evidence regarding the proposed workflows
Iteration	Improved prototype based on testing
Handoff	Documentation sufficient for future software development
Future Roadmap	Recommendation regarding what should be developed next

12. Project Scope

In Scope	Not Committed Within Initial Scope
Understanding the existing final-project process	Production software development
Stakeholder identification	Production authentication system
Student research	Production database architecture
Previous-project student research	Cloud infrastructure
Mentor research	Enterprise-grade file storage
Programme/coordinator research	Production deployment
Existing process and tool audit	Automated notification infrastructure

Historical project/archive analysis	Integration with university systems
Current-state journey/process mapping	Full Moodle replacement
Research synthesis	Full project repository replacement
Pain-point identification	Grading system
Opportunity prioritization	Chat or communication platform
Problem definition	Mobile application development
Product requirements	
Role and permission definition where required	
Information architecture	
User journeys and workflows	
Low-fidelity design	
High-fidelity interactive prototype	
Usability testing	
Product iteration	
MVP scope definition	
Product requirements documentation	
Development handoff	
Future implementation roadmap	
Final project documentation	
Final presentation	

These boundaries may only be revisited if research identifies a strong need and the implementation can realistically fit within the team's available effort.

13. Effort and Duration

Measure	Commitment
Number of Students	2
Hours per Student per Week	4
Duration	20 Weeks

Hours per Student	80 Hours
Weekly Team Capacity	8 Person-Hours
Total Team Effort	160 Person-Hours
Starting Week	9 August 2026
Expected Completion	Around 27 December 2026

14. Project Phases

Phase	Weeks	Focus	Individual Hours per Student	Team Hours
1. Frame & Understand	1–2	Scope, assumptions, stakeholders and research planning	8	32
2. Research	3–7	Student, mentor, programme and current-system research	20	80
3. Synthesize & Define	8–10	Research synthesis, lifecycle mapping and problem prioritization	12	48
4. Product Definition & Design	11–15	Requirements, IA, user flows, wireframes and prototype	20	80
5. Validate & Iterate	16–18	Usability testing, findings and prototype iteration	12	48
6. Finalize & Handoff	19–20	Final documentation, development roadmap and presentation	8	32
Total	20 Weeks		80 Hours	320 Hours

15. Detailed 20-Week Execution Plan

Week	Focus	Main Activities	Expected Output	Hours per Student	Team Hours
1	Project Alignment	Consolidate faculty discussion, define known context, document assumptions and project boundaries	Initial project brief, assumption map	4	16
2	Research Preparation	Stakeholder mapping, existing information review, interview planning	Stakeholder map, research plan, interview guides	4	16
3	Current Student Research	Understand current project-management practices	Student interview notes	4	16
4	Previous Project Research	Understand completed-project experiences, lost information and retrospective problems	Previous-student research evidence	4	16
5	Mentor Research	Understand mentoring workflow, reviews, feedback and visibility	Mentor interview findings	4	16
6	Programme Research	Understand batch-level management, review, completion and archival	Programme workflow findings	4	16
7	Existing-System Audit	Review tools, templates, documents, archives and institutional systems	Current tool and artefact inventory	4	16
8	Research Synthesis	Consolidate evidence and cluster themes	Research themes and pain-point clusters	4	16

9	Current-State Mapping	Map project journey from initiation to archival	Current-state lifecycle / journey map	4	16
10	Problem Prioritization	Assess frequency, impact, stakeholder value and feasibility	Validated problem statement, opportunity priority and success criteria	4	16
11	Product Definition	Define users, roles, key jobs, MVP boundaries and requirements	Initial product requirements and MVP definition	4	16
12	Information Architecture	Define system organization and major workflows	IA and core user flows	4	16
13	Low-Fidelity Design	Explore and compare possible workflow solutions	Low-fidelity prototype options	4	16
14	High-Fidelity Design I	Translate validated workflows into detailed interaction design	High-fidelity core flows	4	16
15	High-Fidelity Design II	Complete and connect major journeys for testing	Test-ready interactive prototype	4	16
16	Validation Cycle 1	Test the prototype with representative users	Usability-testing notes and findings	4	16
17	Product Iteration	Prioritize issues and refine workflows	Revised prototype	4	16
18	Validation Cycle 2	Re-test critical changes and assumptions	Final validation findings	4	16
19	Product Handoff Preparation	Finalize requirements, rules, edge cases and implementation priority	Product specification and roadmap	4	16
20	Final Handoff	Consolidate findings, final prototype, report and faculty presentation	Final proposal package, roadmap and presentation	4	16
			Total	80	320

16. Key Faculty Review Milestones

Review	Timing	Purpose	Key Decision
Research Readiness Review	End of Week 2	Validate research direction and stakeholder access	Are we investigating the correct process and stakeholders?
Problem Definition Review	End of Week 10	Present research evidence and prioritized problems	What problem should the team actually solve?
Prototype Review	End of Week 15	Present proposed MVP and major workflows	Is the prototype suitable for user validation?
Validation Review	End of Week 18	Present testing findings and iterations	Has the concept demonstrated sufficient value and usability?
Final Handoff Review	Week 20	Present complete research, product and roadmap	Should the product move toward development?

17. Two-Person Team Model

The two students will work on a **single common product**.

The project should use **primary ownership with shared participation**, rather than dividing the work into four completely independent sub-projects.

Workstream	Primary Responsibilities
Product Planning & Coordination	Project plan, scope, faculty coordination, weekly tracking, requirements and decision documentation
Student / Project-Team Research	Participant recruitment, student interviews, previous-project research and student workflow synthesis
Mentor & Programme Research	Mentor interviews, programme processes, reviews, governance and cross-project visibility
Product Design & Prototyping	User journeys, information architecture, workflows, wireframes, prototype and usability-testing preparation

Final ownership can be assigned based on the interests and capabilities of the two students.

Shared responsibilities

Activities All Two Members Should Participate In
Important research discussions
Research synthesis
Problem prioritization
Major product decisions
Prototype reviews
Validation interpretation
Faculty reviews
Final recommendations

This structure keeps individual contribution visible without creating four disconnected project streams.

18. Individual Fellowship Contribution Tracking

Each student will maintain an individual weekly contribution record.

Week	Activity	Individual Contribution	Evidence / Output	Hours
1	Project framing	Example: stakeholder mapping	Stakeholder map	4
2	Research planning	Example: mentor interview guide	Interview guide	4
3	User research	Example: conduct and document interviews	Research notes	4
...	...	...	...	...
20	Final handoff	Example: product documentation / presentation	Final deliverable	4
			Total per Student	80

19. Final Deliverables

Category	Deliverables
Discovery & Research	Research plan, stakeholder map, existing-system analysis, research evidence, current-state lifecycle, research findings, validated problem statement
Product Definition	Product scope, user roles, product requirements, prioritized MVP, information architecture, key user journeys and product workflows
Product Design	Low-fidelity explorations, high-fidelity interactive prototype, important product states and scenarios
Validation	Usability-test plan, testing findings, prioritized usability/product issues, iterated prototype
Handoff	Final validated prototype, product specification, MVP implementation priorities, development recommendations, future roadmap, final report and faculty presentation

20. Success Criteria

Success Area	Criteria
Current System Understanding	The team can clearly explain how final projects currently operate from initiation to archival.
Evidence-Based Problem Definition	The proposed problem is supported by stakeholder research rather than only initial assumptions.
Prioritization	The team identifies a manageable and valuable problem instead of attempting to digitize the entire project ecosystem.
Product Definition	A clear and realistic proposed MVP is defined.
Prototype Completeness	Priority workflows can be experienced through an interactive prototype.
User Validation	Representative users are able to test and provide evidence regarding the proposed workflows.
Iteration	Important findings are incorporated into the final proposed product.

Institutional Value	The final concept demonstrates potential value to students, mentors and/or programme stakeholders.
Future Continuity	Another team can understand what should be developed, why and how the system should behave.
Individual Fellowship Evidence	Each team member has approximately 80 hours of traceable contribution.

21. Future Development Path

If the product concept is validated, a later phase may continue through:

Future Stage	Purpose
Validated Product Concept	Current fellowship outcome
Technical Planning	Define architecture, integrations, data and technical feasibility
Software Development	Build the prioritized MVP
Live Pilot	Test with an actual final-project batch
Measurement	Evaluate adoption, workflow improvement and identified success metrics
Programme Adoption	Decide whether the system should become part of the regular PDM process
Future Batch Evolution	Improve the product based on continued usage

This approach allows the current team to reduce product and user risk before significant engineering effort is committed.

22. Conclusion

This proposal positions the PDM Final Project Lifecycle as a structured product discovery, design and validation initiative rather than beginning with a predefined software solution.

The immediate objective is to understand how final projects currently operate, identify the most important problems experienced by students, mentors and programme stakeholders, and determine where digital intervention can create meaningful value.

Over the proposed 20-week period, the four-member team will contribute approximately 80 hours per student, resulting in a total effort of 320 person-hours. The work will progress from research and current-state understanding through problem definition, product design, prototyping, user validation and final handoff.

The expected outcome is a research-backed and validated interactive prototype of the proposed MVP, supported by clear product requirements, user flows, validation findings and an implementation roadmap.

This approach allows the team to focus its available time on reducing uncertainty around the problem and the proposed solution before significant software-development effort is committed.

If the concept demonstrates sufficient value through research and validation, the final output can serve as a strong foundation for subsequent technical development, live piloting and eventual adoption within the PDM programme.